

of participant: 38

Name: Solution

MATH 241

SUMMER 2014

QUIZ#2

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Problem 1. (5 Points)

Consider the following system of linear equations,

$$\begin{aligned}x_1 + x_2 + x_3 &= 3 \\x_1 - x_2 + x_3 &= 1 \\x_1 - x_2 + \alpha x_3 &= \beta\end{aligned}$$

Find all possible values for α and β that the system has (a) a unique solution, (b) infinitely many solutions and (c) no solution.

Problem 2. (5 Points)

Find the solution for the following system of linear equations using Gauss-Jordan elimination.

$$\begin{aligned}x_1 + 2x_2 + 3x_3 + 4x_4 &= 9 \\x_1 + 2x_2 + 3x_3 + 5x_4 &= 10 \\x_1 + 2x_2 + 5x_3 + 2x_4 &= 11\end{aligned}$$

$$1) \quad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & -2 & 1 & 1 \\ 1 & -1 & \alpha & \beta \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \rightarrow R_2 \\ R_3 - R_1 \rightarrow R_3}} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & -2 & \alpha-1 & \beta-3 \end{bmatrix} \xrightarrow{\substack{R_2 \cdot \frac{1}{2} \rightarrow R_2 \\ 2R_2 + R_3 \rightarrow R_3}} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & \alpha-1 & \beta-1 \end{bmatrix}$$

(a) if $\alpha \neq 1$ (b) $\alpha=1, \beta=1$ (c) $\alpha=1, \beta \neq 1$

$$2) \quad \begin{bmatrix} 1 & 2 & 3 & 4 & 9 \\ 1 & 2 & 3 & 5 & 10 \\ 1 & 2 & 5 & 2 & 11 \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \rightarrow R_2 \\ R_3 - R_1 \rightarrow R_3}} \begin{bmatrix} 1 & 2 & 3 & 4 & 9 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & -2 & 2 \end{bmatrix} \xrightarrow{\substack{R_3/2 \rightarrow R_3 \\ R_3 \leftrightarrow R_2}} \begin{bmatrix} 1 & 2 & 3 & 4 & 9 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{matrix} \textcircled{p} & \textcircled{f} & \textcircled{p} & \textcircled{p} \\ \textcircled{1} & & \textcircled{1} & & \end{matrix}$$

$$x_4 = 1 \quad x_3 = 1 + x_4 = 2 \quad x_2 = \alpha \quad x_1 = 9 - 4x_4 - 3x_3 - 2x_2 = 9 - 4 - 6 - 2\alpha = -1 - 2\alpha$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$